

An Explainable Header-Centric Framework for Large-Scale Semantic Table Interpretation and Data Quality Assessment

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Abstract. Semantic Table Interpretation (STI) links tabular data to knowledge graphs, such as DBpedia and Schema.org. Within this framework, Column Type Annotation (CTA) assigns meaningful semantic types to columns, enabling interoperability and consistent mapping to ontology classes and properties. Most existing approaches still treat column headers as secondary signals, relying primarily on cell values, which limits explainability and generalizability in realistic, noisy metadata settings.

We present an explainable, header-centric CTA framework built on *FinalFormat*, a curated set of 39 actionable types designed for downstream data-quality rules. Building on an initial UCI phase of about 1k columns with dictionaries totalling about 2,100 entries (1,800 format terms and 300 abbreviations), we expanded them by roughly 80 percent to approximately 3,800 entries (2,800 format terms and 1,000 abbreviations, respectively). This modest increase enabled over 100-fold growth in practical coverage, from around 1k valid assignments in the initial phase to roughly 103k valid assignments across 118.6k Kaggle+VizNet headers. For every assignment, *SourceKeywords* provide token-level evidence, exposing which header fragments triggered each format.

The 76 Sherlock/Sato semantic types compress into only 15 core FinalFormats, extended by 24 additional domain-specific types, supporting compact yet expressive column-type vocabularies for STI. We evaluate the framework on heterogeneous benchmarks (UCI, Prague, Kaggle, VizNet/Sato) and on the SemTab 2024 metadata-to-KG track targeting DBpedia and Schema.org. On these SemTab tasks, GT-strict results ($\text{Hit}@1/5 = 0.45/0.48$) rise to 0.80/0.83 under human reconciliation. We also introduce *HeadersIQ*, a 0-100 dataset-level data quality metric aligned with SemTab 2024’s IsGold? Track and the KG2Tables evaluation dimensions. Altogether, the framework satisfies SemTab 2025 requirements for noise robustness, alias and acronym resolution, and NIL handling, enabling trustworthy, large-scale, header-centric CTA. All code, dictionaries, models, and benchmark subsets are released for reproducibility and reuse.

Keywords: Semantic table interpretation, column type annotation, knowledge graphs, metadata-to-KG, data quality assessment, table understanding.

1 Introduction

In recent years, the rapid expansion of open data repositories and domain-specific tabular datasets has made semantic table understanding a central challenge for semantic integration and interoperability with knowledge graphs. Most real-world tables remain poorly annotated or lack explicit semantic types, making semantic integration difficult and prone to errors. **Semantic Table Interpretation (STI)**, which links tables to ontologies and knowledge graphs such as DBpedia and Schema.org, addresses this need. Within STI, the **Column Type Annotation (CTA)** subtask assigns meaningful types to table columns, enabling semantic alignment and high-quality metadata.

Student ID	Last Name	FirstName	Age	Country	Height	BirthDate
I345343	white	3	200	USA	145	3/04/2121
J849486	Stewart	Ronald	28	?	170	0/1/2010
J849486	Johnson	Mary	56	Australia	-200	NULL

Fig. 1. Example of Bad Data/Dirty Dataset.

Figure 1 illustrates a realistic “dirty” dataset with duplicates, missing values and obvious type violations. In our framework, such problems are detected and explained through the semantic types inferred from the terms that appear in bold in the headers.

1.1 Historical context and renewed interest

Influential works such as *Sherlock* (2019) [13] discouraged header-centric features, arguing that matching-based approaches (i) recognize only a limited range of types and (ii) are not robust to malformed or context-dependent data, particularly when relying on dictionary or regular-expression matching. This influenced community practice, where unrecognised headers often fall back to primitive types.

Recent findings challenge this view. A survey by Liu et al. (2023) [23] reports that competitive CTA systems increasingly exploit header signals: “*The table header often directly explains the contents of the column. Making full use of the information from the table header can help to find the column type or properties more efficiently*”.

Since 2019, the **Semantic Web Challenges on Tabular Data to Knowledge Graph Matching (SemTab)** [16, 17, 8, 11, 7] reinforced this shift. The 2024 edition [11] introduced a **Metadata-only** track, which promoted header-centric annotation for privacy and scalability, while the **IsGold? Track [1]** emphasized automated **data quality assessment**. **KG2Tables** (Abdelmageed et al., 2025) [2] further highlights data quality as an open gap (diversity, annotation accuracy/clarity, structural coherence).

Notably, *AdaTyper* (2023) [15], from the Sherlock/Sato group, prioritizes header analysis before cell-based fallback, further supporting the value of header semantics.

1.2 Addressing classical critiques

Our approach mitigates limitations noted by Sherlock [13]: dynamic dictionary expansion increases coverage beyond the limited type range observed in matching-based

systems; normalization and abbreviation mappings capture lexical variants that typically break dictionary-based matching; **FinalFormat** types extend beyond simple categories; and ambiguous headers trigger a complementary ML stage with explicit **NIL** labelling, improving robustness in the presence of noisy or context-dependent metadata.

1.3 Approach and Contributions

This work extends a previous header-centric semantic typing and data-quality framework [37], originally applied to 50 UCI datasets with smaller dictionaries. The present version introduces broader benchmarking, expanded type coverage, improved explainability, and a dataset-level quality metric aligned with recent STI benchmarks.

Key Contributions

1. **Broader Benchmarking:** scaling from ~1,000 UCI columns to more than 120,000 across Prague, T2Dv2, VizNet/Sato, Kaggle, SOTAB, and SemTab 2024, annotated with **FinalFormats** and DBpedia/Schema.org properties.
 2. **Dynamic Dictionary Expansion:** a modest 80 percent growth in the dictionaries enabled a more than 100-fold increase in coverage, rising from about 1k to roughly 103k valid assignments across 118k headers.
 3. **Explainability:** **SourceKeywords** provide transparent token-level justifications for every semantic type assignment.
 4. **Expanded Space:** **FinalFormat** grew from 23 to 39 interpretable semantic types.
 5. **Hybrid Machine Learning Pipeline:** five classifiers (Random Forest, Logistic Regression, Gradient Boosting, KNN, Linear SVC) were implemented on 100 data sources (50 UCI + 50 Prague) to complement rule-based detection.
 6. **Benchmark-driven re-evaluation:** reconciliation of SemTab 2024 Metadata-to-KG annotations improves Hit@1/5 from 0.45/0.48 to 0.80/0.83.
 7. **HeadersIQ:** a lightweight dataset-level metric (0–100), enabling scalable, comparable data quality monitoring and aligning with SemTab 2024 “IsGold?”.
 8. **Taxonomy Consolidation:** the 76 Sherlock/Sato labels map deterministically into 15 core **FinalFormat** types. Beyond this, 24 additional domain-specific formats capture long-tail semantic variation, covering ~6 percent of **Universe B** headers (over 7,000 columns), substantially extending the expressiveness for data-quality assessment.
- In summary, the framework provides scalable, explainable, and metadata-only semantic type annotation, aligning with STI benchmarks. All artifacts are publicly available in [38] for reproducibility.

2 Related Work

2.1 Semantic Table Interpretation and CTA Tasks

As outlined in the Introduction, STI links tables to knowledge graphs, and CTA assigns semantic types to columns within this process. Here, we review how existing CTA approaches exploit metadata, especially headers, in STI pipelines.

The SemTab 2024 Challenge has focused on metadata-only annotation.

In particular, the SemTab 2024 Metadata-to-KG track introduced benchmarks where cell values could not be used, reflecting practical concerns about privacy, sensitive data, and large-scale processing. These developments highlight the need for scalable and interpretable methods that rely solely on metadata, such as column headers.

2.2 Column Type Annotation Paradigms

Following Liu et al. [23], CTA systems can be grouped into four paradigms: (i) heuristic and lookup-based methods that rely on string similarity or dictionary matching against vocabularies or KG labels, such as Wang et al. [44], C2 [19], MAGIC [40] and Alobaid et al. [3]; (ii) heuristic and iterative methods that propagate constraints or distribute semantic cues across table elements, including TableMiner+ [47], CSV2KG [41], T2K [34] and the MTab family [29–31]; (iii) feature-based machine learning approaches that use engineered features such as token overlap, datatype patterns and label similarity, for example Limaye et al. [22] and Mulwad et al. [27, 28]; and (iv) deep learning and embedding-based methods, including Sherlock [13], Sato [45], TURL [10], Doduo [42], RECA [39] and DAGOBAB [6], which exploit representations learned from large table corpora. Only a small subset of systems (12 in Liu’s 70 category) deals exclusively with headers, and even fewer operate without access to cell values, leaving an underexplored region in the CTA landscape and motivating our header-centric, metadata-driven design. A broader comparative analysis, including additional recent models, is provided in *Appendix A* [38].

3 Methodology

Figure 2 shows our end-to-end pipeline for semantic type detection and annotation.

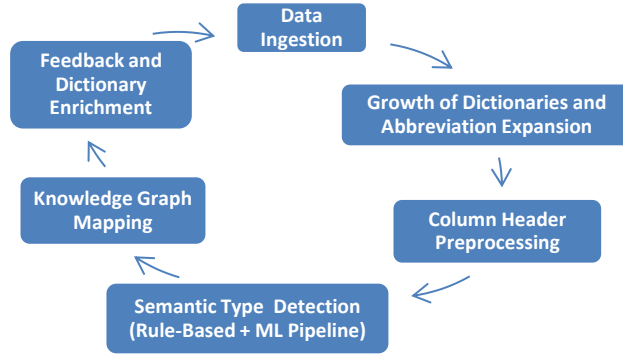


Fig. 2. Pipeline for Semantic Type Detection. The process operates in a header-only mode.

3.1 Data Ingestion

The ingestion stage serves as the entry point for tabular data from diverse sources, including relational databases, spreadsheets, CSV files, and main tables within larger

datasets, such as those from UCI [24] or Kaggle [18]. Because these sources vary widely in structure, ingestion handles multiple file formats and extracts initial metadata such as column headers, optional descriptions and, when available, indicators of Primary or Foreign Keys. Basic validation filters incomplete or malformed tables, and character encodings are normalized to ensure consistent downstream processing. The resulting metadata forms the basis for dictionary expansion and header preprocessing.

3.2 Growth of Dictionaries and Abbreviation Expansion

A key component of the framework is a continuously evolving *formats dictionary* [38] that maps header tokens (including synonyms, variants, and domain-specific terms) to a curated set of **39 FinalFormat** types (see Section 3.4). This resource supports robust, explainable semantic type detection by normalizing diverse column headers and ensuring consistent mapping across heterogeneous domains.

Format Dictionary Creation and Growth. The *formats dictionary* was first developed in previous work [37], derived from header tokens easily associated with semantic formats in the initial datasets. To strengthen coverage and consistency, terms from the Web Data Commons project [21] were incorporated; this corpus analyzed more than 90 million tables in the Web Table Corpus [5] and linked column headers to DBpedia entities via the DBHeaders file [9], yielding an initial dictionary of roughly 1,100 unique header terms. The resource later grew to 1,800 entries in the version reported in [37] (through ChatGPT-based suggestions and manual curation), and subsequently to ~2,800 entries through feedback from approximately 120,000 analyzed columns. During annotation, headers labelled as *NIL* (those lacking a mapping) were reviewed, and their new or variant tokens were added, ensuring continual refinement and broader lexical coverage. The resulting dictionary supports all 39 **FinalFormat** semantic types. *Table 1* illustrates representative mappings.

Table 1. Example mappings from terms to **FinalFormats** (in *formats dictionary* file [38])

Term->FinalFormat type	Term->FinalFormat type
id->IDcolumn	author->name
day and time->datetime	website->URLformat
numeric->numerical	anniversary->date
boolean->binary	day of week->weekday
qualitative->categorical	address->street
height->numerical>= 0	postal code->postalcode
letter->string	e mail->E-mailformat
birthplace->city	internet protocol->IPformat
territory->state	cellphone->phone

These mappings ensure that semantically related or ambiguous headers are consistently normalized to interpretable **FinalFormat** types, such as *name*, *city*, *binary*, *categorical*, and *postalcode*. This capability is essential for handling noisy, abbreviated or variant metadata, as increasingly required by recent STI and SemTab benchmarks.

Abbreviation Dictionary Growth and Expansion. To complement the *formats dictionary*, we maintain a dedicated *abbreviations dictionary* containing over 1,000 mappings from shorthand or compressed forms to their expanded variants, as registered in the *formats dictionary*. This expansion is crucial for decoding terse headers (e.g., “DOB” → *Date of Birth* → *date* or “pct” → *percentage*).

Figure 3 illustrates representative abbreviation categories, covering acronyms, misspellings (e.g., “addre” → *address*, “phn” → *phone*, drawn partly from *NameGuess* [46]), domain jargon and truncations. In every case, at least one term in the expanded form links to a *FinalFormat* type; for example, “ceo” → *chief executive officer* → *name*.

Acronyms			Misspellings		
Abbreviation	Expansion	FinalFormat	Abbreviation	Expansion	FinalFormat
CEO	chief executive officer	name	addre	address	street
DOB	date of birth	date	lettr	letter	string
GPA	grade point average	numerical>=0	phn	phone	phone
Domain jargon			Truncations		
Abbreviation	Expansion	FinalFormat	Abbreviation	Expansion	FinalFormat
bp	blood pressure	bloodpressure	nbr	number	numerical
gp	games played	numerical>=0	mo	month	month
vs	versus	binary	wgt	weight	numerical>=0

Fig. 3. Abbreviation categories and their expansions

Like the main *formats dictionary*, the *abbreviation dictionary* also grows iteratively through *NIL* analysis and expert review, supporting continual adaptation as new datasets and types of noise are encountered.

Alignment with the SemTab 2025 Challenge Priorities. This combined strategy of dictionary growth and abbreviation expansion directly targets the core objectives of the SemTab 2025 challenge [7], specifically, **robust alias/acronym resolution**, **noise resilience**, *NIL* detection for unmappable headers, and ambiguous metadata interpretation.

3.3 Column Header Preprocessing

Preprocessing converts raw column headers into actionable semantic annotations using the *FinalFormat* type system. The main stages are:

1. **Metadata Extraction and Normalization.** Extract header terms and optional descriptions, normalize case and punctuation, and split compound or camelCase tokens to standardize heterogeneous naming conventions.
2. **Abbreviation and Variant Handling.** Expand over 1,000 abbreviations and spelling variants, enabling robust treatment of noisy, truncated or misspelled headers commonly found in real-world data.
3. **SourceKeywords Extraction.** Identify core semantic tokens (e.g., “age”, “id”, “amount”) and contextual cues (e.g., “has”, “is”, “bp”) from header names and descriptions. Unlike prior black-box approaches, the framework explicitly records these *SourceKeywords*, providing a transparent justification for each assignment.

3.4 Semantic Type Detection System

The **FinalFormat** system provides the semantic backbone of the framework, defining 39 interpretable types that sit between coarse atomic datatypes (e.g., string, integer, date) and fine-grained ontology classes in DBpedia or Schema.org. This layer is expressive enough to support meaningful data-quality checks while remaining simple and explainable.

Definition and Scope.

FinalFormat types cover categories such as:

- **Numerical:** generic numeric, non-negative, and bounded numerical values (in blue in **Figure 4**, e.g., percentage, age, bloodpressure, pH, latitude, etc.).
- **Geographical:** city, country, postalcode, state, street.
- **Temporal:** date, datetime, day, hour, month, time, week, weekday, year.
- **Categorical/Qualitative:** Keywords such as “class”, “status”, “type”, etc.
- **Name:** Keywords such as “director”, “actor”, “publisher” or “firstName”.
- **Identifiers:** IDcolumn (e.g. “ISBN”, “SSN”), URLformat, IPformat, phone.
- **Binary:** Boolean fields or fields identified through cues (“has”).
- **Textual:** descriptive, free text, assigned to format string.

#	FinalFormat	#	FinalFormat	#	FinalFormat
1	acidity	14	heartrate	27	percentage
2	age	15	hour	28	ph
3	alkalinity	16	IDcolumn	29	phone
4	angle	17	IPformat	30	postalcode
5	binary	18	latitude	31	saltness
6	bloodpressure	19	longitude	32	state
7	categorical	20	modelname	33	street
8	city	21	money	34	string
9	country	22	month	35	time
10	date	23	name	36	URLformat
11	datetime	24	normalized	37	week
12	day	25	numerical	38	weekday
13	E-mailformat	26	numerical>=0	39	year

Fig. 4. The 39 **FinalFormats**

Matching and annotation.

Semantic type assignment combines deterministic rule-based matching with a robust machine-learning (ML) pipeline to maximise accuracy, coverage and explainability.

Rule-based matching.

Exact, substring and fuzzy matching are applied to assign **FinalFormat** types and link headers to knowledge graph properties, using the curated **formats** and **abbreviations dictionaries**.

Machine-learning pipeline.

For non-ID columns, cleaned and abbreviation-expanded headers (and optional descriptions) are processed by five ML classifiers. Rule-based outputs, such as **SourceKeywords**, are included as explicit features. Headers that cannot be mapped to any **FinalFormat** after all rule-based and ML checks are assigned the value **NIL**, making unmapped cases explicit and available for later review. Implementation details, including preprocessing, feature extraction, model selection and reproducibility settings, are documented in **Appendix B** [38].

3.5 Knowledge Graph Mapping

Each **FinalFormat** type was linked to the most semantically aligned property in DBpedia [4] and Schema.org [20]. Following the semantic annotation method introduced in GitTables [14], both column headers and ontology labels were converted into dense character-level n-gram vectors using pretrained FastText embeddings, and cosine similarity between these representations was used to select the most semantically aligned property/class. This enabled robust matching of lexical variants and abbreviations, providing a more reliable alignment than exact string matching.

Figure 5 illustrates some real mappings, showing the full pipeline of header normalization, abbreviation expansion, dictionary matching, **FinalFormat** assignment, and **SourceKeyword** extraction, alongside ontology alignment results. The examples highlight both accurate matches and areas of limited ontology coverage, for instance, *dateOfAbandonment* and *participant* in **DBpedia**, or several **Schema.org** terms that were incorrectly aligned despite high similarity scores (shown in red). These issues reveal that although KG-based annotation benefits from abbreviation expansion (e.g. *dob* and *wt*), semantic accuracy is not always guaranteed.

Column	Source Keyword	FinalFormat	DBpediaType	DBpedia Score	SchemaOrgType	Schema Score
Birth year	year	year	birthYear	1	birthDate	0.77
dob	date of birth	date	dateOfAbandonment	0.95	releaseOf	0.87
employee type	type	categorical	type	1	employee	1
line size	size	numerical>=0	collectionSize	0.81	line	1
participant age	age	age	participant	1	participant	1
wt	weight	numerical>=0	weight	1	weight	1

Fig. 5. Semantic Type Assignment for **FinalFormat** and KGs

Overall, **FinalFormat** delivers finer-grained classifications than ontology mappings or atomic datatypes, enhancing semantic interpretability and data-quality profiling.

3.6 Feedback and Dictionary Enrichment

NIL-labelled headers are reviewed through a controlled, evidence-based human-in-the-loop feedback cycle that evolves with each benchmark iteration.

NIL Token improvement process.

All columns with **FinalFormat** labelled as **NIL** follow this protocol:

- Aggregate and rank all unmatched tokens by frequency.
- Filter out meaningless or unstable entries.
- Manually review high-frequency candidates for domain relevance.
- Integrate only validated and interpretable terms into the **Formats** and **Abbreviations Dictionaries**.

3.7 Integration with Data Quality Assessment

As detailed in previous work [37], the framework links semantic type assignments to automated data quality rules. Each *FinalFormat* triggers specific quality checks (e.g., uniqueness, format validation). The *HeadersIQ* metric summarizes the overall dataset quality (see Section 4.8), aggregating all detected **Data Quality Issues (DQIs)** into a single interpretable score, supporting large-scale, automated profiling.

4 Results and Coverage Analysis

This section presents a comprehensive evaluation of the header-centric semantic type detection framework across a diverse suite of tabular data benchmarks, including both classical STI datasets and large, real-world sources. We report coverage and accuracy for *FinalFormat* assignments and for knowledge graph mappings (DBpedia [4] and Schema.org [20]). Qualitative results, including interpretability through *SourceKeywords*, complement the quantitative analysis.

Sherlock [13] highlighted two main limitations of matching-based, header-dependent systems: they tend to recognize only a limited range of types, and they lack robustness when confronted with noisy or context-dependent attributes. The evaluation presented in this section directly addresses both points. Broad lexical coverage is demonstrated through the *dictionaries (formats and abbreviations)* and the 39 *FinalFormat* types. Robustness is demonstrated through consistently high assignment rates across heterogeneous and noisy repositories, such as Kaggle and VizNet, where normalization, abbreviation expansion, *SourceKeywords*, and *NIL*-driven feedback enhance resilience under conditions where matching-based systems have previously been shown to fail.

4.1 Datasets and Benchmarking Scope

Scalability, robustness and generalizability were evaluated across seven heterogeneous benchmarks (*Table 2*). The evaluation spans structured repositories (UCI [24], **Prague Relational Learning Repository** [26]), large sets of web tables (**Sato/VizNet** [45, 12]), user-generated real-world tables (**Kaggle** [18]) and KG-grounded benchmarks (**T2Dv2** [35] (* denotes that the 39 “Areas” correspond to DBpedia classes), **SOTAB** [20]). *Table 2* reports structural statistics for all benchmarks.

Table 2. Structural Statistics of Benchmarks

Benchmark	Domains	Datasets	Columns	Columns per Dataset
UCI	6	50	922	18.4
Prague	9	50	520	10.4
Sato	N/A	2,254	4,587	2
VizNet	N/A	11,199	74,915	6.7
Kaggle	17	3,364	49,727	14.8
SOTAB	N/A	N/A	737	N/A
T2Dv2	39*	237	1,172	4.9

Universe definitions.

- **Universe A** - All seven benchmarks.
- **Universe B** - Kaggle + VizNet (118,639 columns). Used to study distribution, long-tail behaviour and bounded-numerical coverage at scale.

4.2 Semantic Type Detection: Coverage and Diversity

We measure three aspects of semantic coverage across all benchmarks:

- **FinalFormat Types**: number detected in each dataset group.
- **SourceKeywords**: **unique, normalized header tokens** extracted across all.
- **DBpedia/Schema.org Types**: number of **distinct ontology classes/properties** successfully linked from column headers to the respective ontologies after normalization and abbreviation expansion.

Table 3 summarizes results. **VizNet** and **Kaggle** display much higher **SourceKeywords** due to scale, yet headers are still resolved into our **39 FinalFormats** for Kaggle.

Table 3. Coverage of *FinalFormat*, *SourceKeywords*, and KG Types

Bench- mark	<i>FinalFormat</i> Types	<i>Source Key- words</i>	DBpedia Types	Schema.org Types
UCI	33	248	162	161
Prague	23	139	136	129
Sato	15	76	76	63
VizNet	35	1,792	1,045	1,010
Kaggle	39	1,920	1,046	1,138
SOTAB	23	349	246	684
T2Dv2	21	234	213	172

A key result is that **only 15 core FinalFormats** deterministically cover the entire **76-label Sherlock/Sato space** in many-to-one mappings. Two extra labels from Sherlock do not appear in Sato (*File size* and *Team name*). The consolidation preserves distinctions important for data quality (e.g., IDcolumn, date, postcode), and redundant semantic variants (e.g., creator-type labels) merge naturally into compact categories, such as name or categorical.

Within **Universe B (Kaggle + VizNet)**, these 15 formats alone cover **~80%** of all 118,000 columns. The distribution is heavily concentrated in a few operationally critical types (categorical, numerical ≥ 0 , IDcolumn, string, numerical). The complete list of these 15 types and their mappings to Sherlock/Sato is shown in **Appendix C** [38].

Beyond the 15 core types, **Universe B** exhibits an additional **24 FinalFormats**, primarily bounded numerical values and domain-specific formats such as *percentage*, chemical (*ph*, *acidity*), health (*heartrate*), geography (*latitude/longitude*), among others. These collectively represent **~6%** of columns, **totaling over 7,000 columns in this universe, which is quite substantial**, and enable finer-grained DQI checks not supported in the Sherlock/Sato taxonomy. The full list is also reported in **Appendix C** [38].

Across *Universe B*, *NIL* ~13 percent (thus *Valid* ~87 percent). This residual corresponds to noisier or novel multi-word headers that did not match any of the *FinalFormats* after normalization and abbreviation expansion.

Scalability. The modest increase of 80% from ~2,100 dictionary entries (UCI phase) to ~3,800 entries supported a growth of ~103 times in practical coverage, increasing valid assignments from ~1,000 to ~103,000 columns. Details appear in *Table 4*.

Table 4. Scalability evidence

Phase	Dictionary size	Columns evaluated	Valid assigned
UCI baseline[37]	1,800 + 300 $\approx$ 2,100	~1,000	~1,000 (near perfect)
Kaggle+VizNet	2,800 + 1,000 $\approx$ 3,800	118,639	~103,000 (~87%)

Bounded numerical formats. *Appendix D* [38] lists all adopted bounds (e.g., age 0–130, ph 0–14). These constraints support DQI checks and help disambiguate numeric families within the *FinalFormat* taxonomy. They are based on subjective design choices that can be adjusted easily as domain requirements evolve.

4.3 *NIL* Token Improvement Process

A qualitative review of the top 100 *NIL* cases in *Universe B* shows that most unresolved headers are extremely short codes or symbols (e.g., *no*, *so*, *ga*, *po*) that carry little or no semantic information and therefore fall outside any meaningful *FinalFormat* mapping.

A smaller subset of cases (e.g., *streak*, *medal*, *designation*, *msrp*) exhibits domain-specific or potential abbreviations that may justify future additions to the dictionaries.

These observations confirm that the remaining *NIL* proportion primarily reflects low-information tokens or highly specialized terminology rather than systematic gaps. The complete ranked list of 100 *NIL* tokens is included in *Appendix E* [38].

4.4 Validity, *NIL* Rates, and Robustness

Table 5 reports *NIL* counts and Valid percentages for (i) *FinalFormat* assignments and (ii) subsequent mappings to DBpedia and Schema.org across *Universe A*.

Table 5. *NIL*/Valid Assignment Rates

Benchmark	Final Format <i>NIL</i>	Final Format Valid %	DBpedia <i>NIL</i>	DBpedia Valid %	Schema.org <i>NIL</i>	Schema.org Valid %
UCI	12	98.7	433	53	413	55.2
Prague	0	100	94	81.9	93	82.1
Sato	0	100	0	100	423	90.8
VizNet	9,435	87.4	21,244	71.6	23,795	68.2
Kaggle	7,274	85.4	16,546	66.7	17,467	64.9
SOTAB	5	99.3	163	77.9	7	99.1
T2Dv2	267	77.2	305	74	469	60

Definition of *NIL* and *Valid%*: *NIL* denotes the number of cases where no *FinalFormat* match is found after all rule-based and ML checks. *Valid%* is computed as $(1 - \text{NIL} / \text{Total Columns}) \times 100$.

For DBpedia and Schema.org, *Valid%* measures the fraction of *FinalFormat* types that could be aligned with at least one ontology property or class.

Key Observations.

- DBpedia and Schema.org mappings remain robust, with most *FinalFormat* types finding at least one compatible ontology concept.
- Sato and Prague obtain 100 percent *FinalFormat Valid*, confirming the completeness of the dictionaries and rules for structured benchmarks and showing that the framework fully subsumes the Sherlock/Sato label space.
- UCI and Kaggle exhibit lower KG mapping coverage, reflecting the presence of heterogeneous, noisy, or domain-specific headers not represented in current ontologies.

4.5 Machine Learning Semantic Type Detection System

In addition to the rule-based pipeline, we evaluated a supervised machine-learning component trained on the 50 UCI datasets and 50 Prague relational tables. Five classifiers (Random Forest, Gradient Boosting, KNN, Logistic Regression, Linear SVC) were tested using a balanced and fully validated corpus of approximately 1,400 columns. Several formats with clear lexical signatures (e.g., URLformat, E-mailformat, Age, Date, City, Country) achieved **near-perfect** precision, recall, and F1 scores across all models.

By contrast, the large-scale rule-based evaluation on 118,639 Kaggle+VizNet columns yielded a *NIL* rate of approximately 13 percent, driven by ambiguous, noisy, or domain-specific headers typical of open-domain data. This clarifies that the ML metrics represent an upper-bound scenario with clean supervision. In contrast, the large-scale *NIL* proportion represents real-world variability rather than limitations of the machine-learning models themselves. Complete per-format metrics appear in *Appendix B* [38].

4.6 Handling Complex and Ambiguous Header Cases

Beyond straightforward headers, the system also resolves complex, abbreviated, and domain-specific cases through the combined effect of normalization, abbreviation expansion and *SourceKeywords* extraction. These mechanisms enable the framework to interpret signals such as “is”, “has”, “or not” (binary cues), “range” (categorical context), “birth place” (geographical semantics) or medical abbreviations such as “trestbps” (bloodpressure). This avoids misclassification in cases where multiple semantic cues coexist or where the lexical form is highly compressed.

These challenging cases demonstrate the system’s ability to surpass surface-level matching and produce interpretable, rule-aligned *FinalFormats*, even in noisy or unconventional metadata. A detailed description of the rule-based preprocessing, normalization, abbreviation expansion and KG annotation steps, together with an extended set

of worked examples showing how *SourceKeywords* and *FinalFormats* are assigned to complex headers, is provided in *Appendix F* [38].

4.7 Comprehensive Benchmark Results on the SemTab 2024 Metadata-to-KG Track

Ground-truth imperfections. As detailed in *Appendix G* [38], the ground truth for the SemTab 2024 Metadata-to-KG track contains typical large-scale benchmark issues, including duplicates, label-granularity inconsistencies, and occasionally misassigned entities. In line with previous SemTab practices, we first report the official *GT-strict* scores and then present our internal audit, which is applied exclusively to our own predictions.

Official (GT-strict) results. Using the organizers’ ground truth and script, the system correctly annotated **64 of 141 columns (Hit@1 = 0.45)** and **68 of 141 (Hit@5 = 0.48)**. Baseline scores for Adwan [43] and CVA/MetaLinker [25] are taken from the official SemTab 2024 report’s Table 3 for Round 1 [11]. Panel A of *Table 6* presents these figures.

Internal audits of GT (ours only). We performed a blinded, post-hoc inspection of our predictions (*Appendix G*), and each was reassessed under two diagnostic categories:

1. **Correct** - the prediction matches the intended GT interpretation, even where the published GT label is noisy or inconsistent.
2. **Appropriate** - the prediction is semantically reasonable but differs from the GT label due to synonymy, granularity, or benchmark inconsistencies.

Panel B of *Table 6* summarises these results. These scores are *not* used for comparison with other systems; they quantify benchmark noise and semantic appropriateness.

Metric note. NIL denotes cases with no match after all normalization and abbreviation expansions have been applied. Valid is defined as $1 - \text{NIL}/\text{total}$. All NIL cases count as errors in Hit@k.

Table 6. Comparative analysis on the SemTab 2024 Metadata-to-KG track

Panel A - Official (GT-strict)		
System	Hit@1	Hit@5
Ours (GT-strict)	0.45	0.48
Adwan [43]	0.75	0.92
CVA / MetaLinker [25]	0.55	0.70
Panel B - Human-reconciled (ours only)		
Category	Hit@1	Hit@5
Correct only	0.76	0.79
Correct + Appropriate	0.80	0.83

4.8 *HeadersIQ*: Dataset-Level Data Quality Measurement

We define *HeadersIQ* as a simple dataset-level score that estimates the proportion of cells that comply with our data quality rules. Let T be the number of cells checked and E the number of detected Data Quality Issues; then $\text{HeadersIQ} = (1 - E/T) \times 100$. Scores close to 100 indicate no observed problems, whereas lower values reflect increasing levels of intrinsic and representational noise. This compact 0–100 metric enables us to compare tabular sources of varying sizes to identify datasets, aligning with the **SemTab 2024 “IsGold?” track objective** and the **KG2Tables** framework. Representative examples and the full table of results are provided in *Appendix H*, along with implementation details from the *Attribute-BasedDataQualityAssessment.ipynb* notebook.

5 Discussion

5.1 Reassessing Header-Centric Type Detection

Header-only annotation has often been regarded as unreliable, partly influenced by early approaches such as Sherlock [13] and Sato [45]. Our results challenge this view. With systematic preprocessing, curated and incrementally enriched dictionaries, abbreviation expansion, and a complementary ML stage, headers alone contain enough semantic structure to support precise and scalable type assignments. The 39 *FinalFormat* types form a middle layer between primitive datatypes and fine-grained ontology classes, being expressive enough for KG mappings yet simple enough to trigger interpretable data quality rules.

5.2 Scalability, Coverage and Benchmark Alignment

A modest 80 percent expansion of the dictionaries enabled more than 100-fold growth in coverage, rising from approximately 1,000 valid assignments in the UCI phase to roughly 103,000 across more than 118,000 heterogeneous headers. This demonstrates that NIL-driven feedback scales lexically without requiring a significant growth in the dictionaries. At the same time, the 76 Sherlock/Sato types map cleanly into only 15 core *FinalFormat* categories. Additionally, 24 domain-specific types extend the taxonomy to cover real-world long-tail cases without creating an overly fragmented label space. Consistent results across UCI, Prague, Sato/VizNet, SOTAB and T2Dv2 indicate that the same 39 types are sufficient to support CTA pipelines that map columns to DBpedia and Schema.org in a uniform way.

5.3 Complementary ML stage and Ambiguity Handling

The ML component performed almost perfectly under noise-free conditions on 100 curated data sources, but its role remains secondary at scale. It primarily disambiguates

difficult headers and supports rare types, complementing the ~ 13 percent *NIL* rate observed on Kaggle and VizNet. This suggests that small, well-targeted models are sufficient when explainable rules resolve most assignments.

5.4 Dataset-Level Quality via HeadersIQ

HeadersIQ provided stable and intuitive quality scores across datasets of varying sizes, offering a scalable method for monitoring reliability in large data repositories. Its design aligns closely with the SemTab 2024 “IsGold?” emphasis on dataset-level reliability, demonstrating the practical value of linking semantic typing with automated data-quality assessment.

5.5 Interpreting Ground Truth in SemTab 2024

Official SemTab 2024 Metadata-to-KG results yielded $\text{Hit}@1 = 0.45$ and $\text{Hit}@5 = 0.48$. A blinded, header-only audit showed that many strict “errors” reflect GT granularity or aliasing effects; the diagnostic $\text{Hit}@1$ reached ~ 0.80 when semantically appropriate refinements were included. This highlights that rigid GT matching underestimates realistic header-centric correctness and suggests future CTA benchmarks should incorporate multi-label or graded evaluation and explicit human-in-the-loop validation.

5.6 Limitations and Future Work

The main limitations are (i) dependence on meaningful headers and (ii) difficulty with long-tail domain terms. Future work should combine header and value signals, integrate multilingual models, and explore LLM-based disambiguation for sparse or overloaded metadata.

6 Conclusion

This paper presents an explainable, header-centric CTA framework that operates in metadata-only settings by combining normalization, curated dictionaries, *SourceKeywords* and a complementary ML stage to assign 39 *FinalFormat* types. The same abstraction layer supports mappings to DBpedia and Schema.org, in addition to driving a rule-based data quality pipeline, yielding a single dataset-level score (*HeadersIQ*) that summarises intrinsic and representational quality.

Experiments on more than 120,000 columns from seven benchmarks, along with an audit of the SemTab 2024 Metadata-to-KG track, demonstrated high coverage, full compatibility with the Sato/Sherlock label space, and a strong alignment between header-centric annotations and human judgment. This positions the framework as a practical bridge between large tabular data collections, KG-oriented benchmarks and dataset-level quality assessment and opens opportunities for multilingual and LLM-assisted extensions in future work.

Disclosure of Interests.

The authors have no competing interests to declare that are relevant to the content of this article.

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The authors utilized AI-assisted tools (OpenAI ChatGPT) to support coding tasks and assist in drafting, refining, and clarifying explanatory text throughout the manuscript, including detailed narrative sections such as the Metadata-to-KG analysis. The AI-assisted coding also produced derived analysis files (e.g., intermediate annotations, processed columns, and diagnostic outputs), but no synthetic data was generated. No AI system contributed to scientific content, research ideas, methodological design, experimental protocols, quantitative analysis, interpretation of results, or the creation of novel methods. All scientific contributions are entirely the authors' own.

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